

Automated Test Report

29-12-2025

DUT Name	BBNL MPPT 1kW – [MPPT DC/DC Converter]
DUT Part number	TPS1030004A
DUT Rev & Checksum	S04 – FFABD
Testing person	VISHNU C V

Test Report Summary

SI No	Test Name	Result
1	Input voltage range verification	Fail
2	Output voltage accuracy	Fail
3	Load regulation	Pass



PowerSafe-X

Test Item 2: Output voltage accuracy

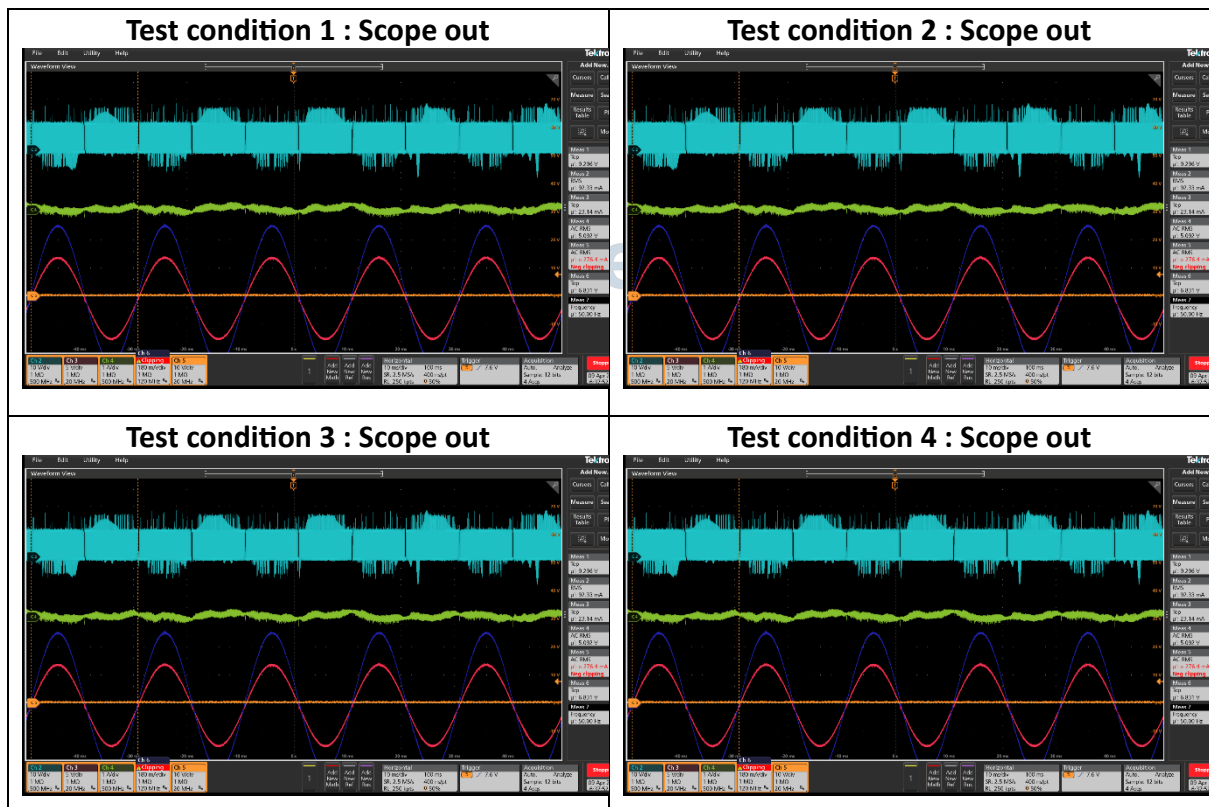
Verification of Input voltage range verification

Test Spec:

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



Timestamp: XXXXXXXX

Test Result: **Fail**

Test Item 3: Load regulation

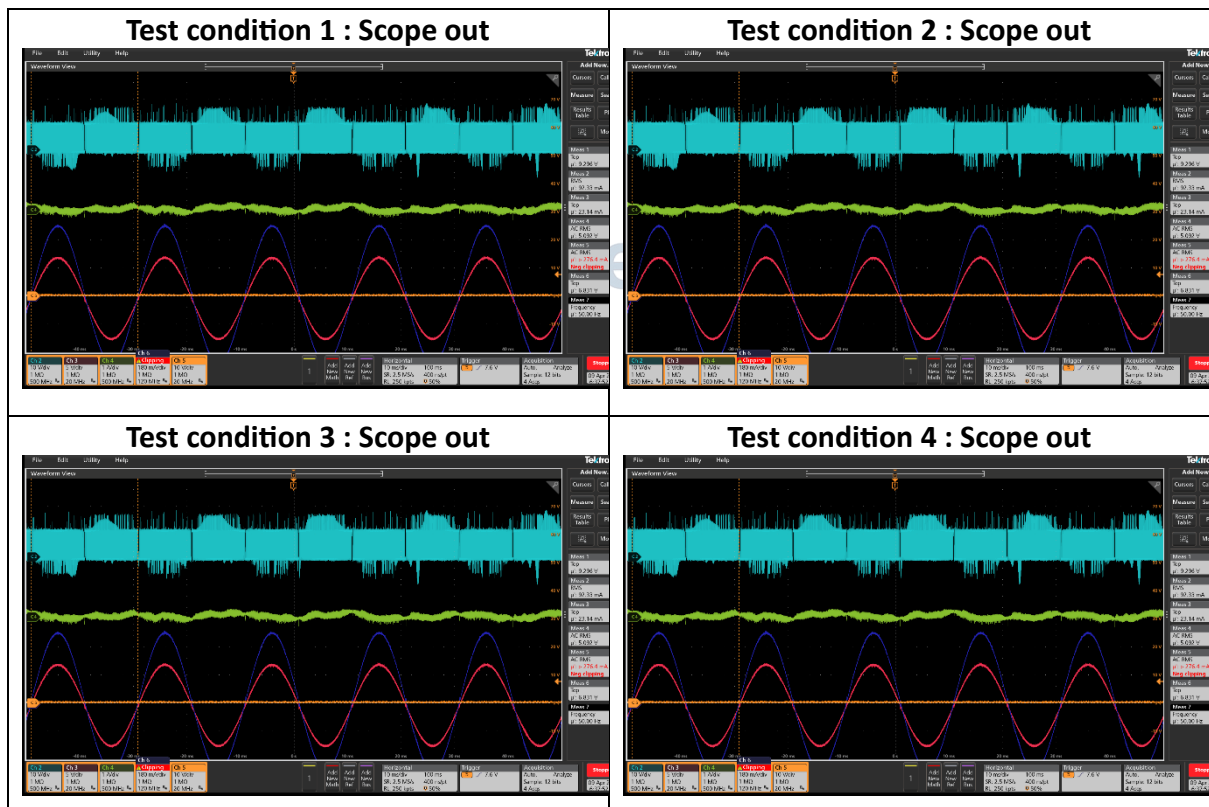
Verification of Input voltage range verification

Test Spec:

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



Timestamp: XXXXXXXX

Test Result: **Pass**

Test log

[13:55:37] Initializing Instruments...
[13:55:37] --- Starting Test: Input voltage range verification ---
[13:55:37] Running Global Setup (130% safety limits)...
[13:55:37] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[13:55:38] Setting Load Limits: I_{max}=23.85A
[13:55:38] Step 1: Setting Source Voltage to Nominal (80.0V)
[13:55:38] Step 2: Load Config -> Mode: CC, Value: 2.0
[13:55:38] Step 3: Turning Source ON
[13:55:38] Step 4: Waiting for V_{out} to reach 95% threshold (51.77V)...
[13:55:47] Timeout waiting for condition after 10s
[13:55:47] Error: Timeout waiting for V_{out} to reach 95% nominal
[13:55:47] Capturing screenshot: D:\Vishnu\Antigravity Workspace\Test Automation GUI\reports\captures\20251229_135547_Input_voltage_range_verification.png
[13:55:53] --- Starting Test: Output voltage accuracy ---
[13:55:53] Running Global Setup (130% safety limits)...
[13:55:53] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[13:55:53] Setting Load Limits: I_{max}=23.85A
[13:55:53] Step 1: Setting Source Voltage to 48.0V
[13:55:53] Step 2: Turning Source ON
[13:55:54] Step 3: Measuring output voltage at no load
[13:55:54] No Load V_{out}: 24.002V
[13:55:54] Error: 55.96%
[13:55:54] Step 4: Testing at 25% load (4.59A)
[13:55:56] V_{out}: 23.921V, Error: 56.11%
[13:55:56] X FAIL: Exceeds 5.0% tolerance
[13:55:56] Step 5: Testing at 50% load (9.17A)
[13:55:58] V_{out}: 23.931V, Error: 56.09%
[13:55:58] X FAIL: Exceeds 5.0% tolerance
[13:55:58] Step 6: Testing at 75% load (13.76A)
[13:56:00] V_{out}: 23.937V, Error: 56.08%
[13:56:00] X FAIL: Exceeds 5.0% tolerance
[13:56:00] Step 7: Testing at 100% load (18.35A)
[13:56:02] V_{out}: 24.072V, Error: 55.83%
[13:56:02] X FAIL: Exceeds 5.0% tolerance
[13:56:02] X OVERALL FAIL: Output voltage accuracy out of specification
[13:56:02] Capturing screenshot: D:\Vishnu\Antigravity Workspace\Test Automation GUI\reports\captures\20251229_135602_Output_voltage_accuracy.png
[13:56:10] --- Starting Test: Load regulation ---
[13:56:10] Running Global Setup (130% safety limits)...
[13:56:10] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[13:56:10] Setting Load Limits: I_{max}=23.85A
[13:56:10] Running Load regulation...
[13:56:12] Capturing screenshot: D:\Vishnu\Antigravity Workspace\Test Automation GUI\reports\captures\20251229_135612_Load_regulation.png

